

ANA SOFIA SILVA PINTO

INFORMATICS AND COMPUTING ENGINEERING

📍 Porto, Portugal

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Informatics and Computing Engineering graduate with a passion for building secure, resilient, and future-proof systems. I thrive in full-stack and backend development, with a strong interest in the challenges at the intersection of cybersecurity and AI.

I'm currently deepening my expertise in modern frontend frameworks like React and TypeScript, and I'm eager to join a forward-thinking team where I can contribute to impactful projects and continuously learn

EDUCATION

COMPLETED STUDIES

Bachelor in Informatics and Computing Engineering

Faculty of Engineering, University of Porto (FEUP&FCUP) | September 2021 - July 2025

- Relevant coursework: Algorithms, Software Engineering, Web Development, Databases, IA, Parallel and Distributed Computing, Computer Networking

WORK EXPERIENCE

SOFTWARE DEVELOPMENT INTERN

Visma Tech Portugal - Porto, Portugal

July 2025 - August 2025

- Contributed to the development of VisionPlanner, a SaaS product for accounting, by implementing backend features in .NET (C#) and integrating frontend components in React.
- Delivered two real user stories, including a critical bug fix for access control and a new feature for automatic user role assignment across distributed services.
- Participated in Scrum ceremonies (planning, dailies, reviews), wrote unit tests, and collaborated via Git, Azure DevOps, and Jira in a professional agile environment.

TECHNICAL SKILLS

- Languages:** C/C++, Java, Python, C#, TypeScript, JavaScript, PHP, Dart, Haskell
- Frameworks & Libraries:** .NET, React, Flutter
- Backend & Databases:** Microservices, Distributed Systems, REST APIs, MySQL, Docker
- Tools & Methodologies:** Git, Azure DevOps, Jira, CI/CD, Agile/Scrum
- Areas of Interest:** Cybersecurity, Artificial Intelligence
- Spoken Languages:** Portuguese (*Native*), English (*Fluent*), Spanish (*Basic*)

PROJECTS

SprintHive - a Project Management Platform (*PHP, JavaScript, MySQL*)

- Developed a **full-stack project** management platform with PHP (Laravel), JavaScript, and MySQL to streamline team collaboration and task management.
- Designed features for **project creation, task assignment, and team communication**, applying **agile methodologies** throughout the development lifecycle.

Java-- Compiler (*Java, ANTLR, Jasmin, JVM*)

- Built a **full, three-stage compiler from scratch** for a Java subset, managing the entire pipeline from front-end lexing/parsing (ANTLR) and intermediate representation (OLLIR) to back-end Jasmin bytecode generation.
- Implemented rigorous semantic analysis** to guarantee code correctness and applied optimizations to generate efficient, executable bytecode for the JVM.

Distributed Chat System (*Java, LLM*)

- Architected a **high-concurrency, secure chat system** leveraging Java 21 virtual threads for optimal performance.
- Secured communications** using SSL/TLS encryption (javax.net.ssl) and ensured robust, thread-safe operations with ReentrantLocks.
- Integrated a local LLM** (via Ollama) to create AI-powered chat rooms capable of contextual, automated replies.

ADDITIONAL INFORMATION

More projects can be seen on my [Github](#) page.

- Interests:** Cybersecurity challenges (CTFs), AI, inspired by music + pop culture.